# Real Analysis 2

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# Contents

# Chapter 1

# Differentiation

## 1.1 Differentiability

**Definition.** Let $f : R^n \to R^m$. We say that $f$ is **Gateaux-differentiable** at $p \in R^n$ when there is a **linear function** $f'(p) : R^n \to R^m$ such that,

$$\lim_{t \to 0} \frac{f(p + tu) - f(p)}{t} = f'(p)u$$

for any $u \in R^n$

**Remark.** This is basically just the directional derivative with a linear function as the derivative at a specif

**Definition.** We say $f$ is **Frechet-differentiable** at $p$ when $f$ is Gateuax-differenitable at $p$ and,

$$f(p + u) = f(p) + f'(p)u + \varepsilon(u) \, ||u||$$

where $\varepsilon : R^n \to R$ satisfies,

$$\lim_{u \to 0} \varepsilon(u) = 0$$

**Remark.** The point here is that in all directions the error uniformly goes to zero.

**Remark.** In 1D, they are both the same.

**Example.** Consider,

$$f(x, y) = \begin{cases} \frac{x^3}{x^2+y^2} & \text{at } (x, y) \neq 0 \\ 0 & \text{at } (x, y) = 0 \end{cases}$$

Now note we have the derivative at $(x, y) = 0$ as,

$$\begin{aligned} f'(0,0)(a,b) &= \lim_{t \to 0} \frac{f((0,0) + t(a,b))}{t} \\ &= \lim_{t \to 0} \frac{f(ta, tb)}{t} \\ &= \frac{1}{t} \frac{(ta)^3}{(ta)^2 + (tb)^2} \\ &= \frac{t^2 a^3}{(ta)^2 + (tb)^2} \\ &= \frac{a^3}{a^2 + b^2} \end{aligned}$$

however note that the derivative is not linear in the direction. ◇

**Example.** Consider,

$$f(x,y) = \begin{cases} \frac{x^3 y}{x^2+y^2} & \text{at } (x,y) \neq 0 \\ 0 & \text{at } (x,y) = 0 \end{cases}$$

We have,

$$\begin{aligned} f'(0)(a,b) &= \lim_{t\to 0} \frac{f(ta,tb) - f(0)}{t} \\ &= \lim_{t\to 0} t \frac{a^3 b}{a^2+b^2} \\ &= 0 \end{aligned}$$

Which is linear in direction. So it is Gateaux differentiable, now note for Frechet differentiable we have,

$$\begin{aligned} \lim_{(a,b)\to 0} \frac{f(a,b)}{||(a,b)||} &= \lim_{(a,b)\to 0} \frac{a^3 b}{a^2+b^2} \frac{1}{\sqrt{a^2+b^2}} \\ &= \lim_{(a,b)\to 0} \left| \left( \frac{a^2}{a^2+b^2} \right) \left( \frac{a}{\sqrt{a^2+b^2}} \right) b \right| \\ &\leq \lim |b| = 0 \end{aligned}$$

◇

**Theorem 1.1.** If $f$ is differentiable at point $c$ then $f$ is continuous at that point as well.j

**Proof.** Let $f$ be differentiable at $c$, by definition we have, $\forall \varepsilon > 0$ exists $\delta$ such that if $|x-c| < \delta_0$ then we have,

$$\left| \frac{f(x) - f(c)}{x - c} - f'(c) \right| < \varepsilon$$

Now expanding this out we have,

$$\begin{aligned} |f(x) - f(c) - f'(c)(x-c)| &< \varepsilon |x-c| \\ |f(x) - f(c)| &< \varepsilon |x-c| + |f'(c)(x-c)| \\ &< \varepsilon |x-c| + |f'(c)||(x-c)| \\ &< |x-c|(\varepsilon + |f'(c)|) \end{aligned}$$

Now simply choose $\varepsilon = 1$ and we have,

$$|f(x) - f(c)| < |x-c|(1 + |f'(c)|)$$

Now note that we simply can choose $\delta$ such that $|x-c| < \delta = \min\{\delta_0, \frac{\varepsilon}{1+|f'(c)|}\}$ for any choice of $\varepsilon$ and we have $\forall \varepsilon > 0$ a $\delta$ such that,

$$|f(x) - f(c)| < |x-c|(1 + |f'(c)|) < \varepsilon$$

which by definition mean that $f$ is continuous at $c$. □

**Note.** Note that the converse is not true that if a function is continuous it is differential. Take for example $f(x) = |x|$

## 1.2 Mean Value Theorems

Let $f : A \to B$ and $f$ be continuous. Then if $A$ is compact, so is $f(A) \subseteq B$, and if $B \subseteq R$, then $f$ has a max and min on $A$.

**Theorem 1.2** (Interior Maximum Theorem). Let $A \subseteq R^n$. Assume that $p \in int(A)$ is a local max point for $f : A \to R$. Then if $f$ is Gateaux differentiable at $p$, then,

$$f'(p) = 0$$

**Proof.** First note that,

$$f'(p)u = \lim_{t \to 0^+} \frac{f(p+tu) - f(p)}{t}$$

And as $f(p)$ is the max then $f(p+tu) \leq f(p)$ and $f(p+tu) - f(p) \leq 0$ which gives us $f'(p)u \leq 0$
Simlalrly we have,

$$f'(p)u = \lim_{t \to 0^-} \frac{f(p+tu) - f(p)}{t}$$

And here we have $f'(p)u \geq 0$. Both these give us,

$$f'(p) = 0$$

□

**Theorem 1.3** (Rolle's Theorem). Suppose that $f$ is continuous on $[a, b]$ and derivative exists in $(a, b)$ and that $f(a) = f(b) = 0$. Then there exists a point $c$ in $(a, b)$ such that $f'(c) = 0$.

**Proof.** Case 1: $f$ is identically zero in it's domain. In this case for all points in $c \in (a, b)$ we have $f'(c) = 0$

Case 2. If $f$ is not identically zero, there there exists some value in $(a, b)$ that is either smaller or greater than 0. Now clearly $[a, b]$ is compact and hence a continuous function $f$ attains it's maximum or minimum within the set. We also established that $f(a)$ and $f(b)$ cannot be the maximum and minimum in this case. Hence, there must exists some $c \in (a, b)$ which is either the local maxima or the local minima. In either of the cases we have $f'(c) = 0$ □

**Theorem 1.4** (Mean value theorem). Let $f$ be a real valued function continous on $[a, b]$ and differentiate on $(a, b)$. Then, there is a $c \in (a, b)$ such that,

$$\frac{f(b) - f(a)}{b - a} = f'(c)$$

**Proof.** First we'll construct $h$ which is the slope of the points connecting $a$ and $b$. Consider $h(x) = mx + p$. So we need $h(a) = f(a)$ and $h(b) = f(b)$. So we have, $h(a) = ma + p = f(a)$ and $mb + p = f(b)$ which give us $m = \frac{f(b)-f(a)}{b-a}$ and $p = f(a) - ma$.

Now consider $g(x) = f(x) - h(x)$ and note that $g(a) = g(b) = 0$. Clearly we can use Rolles theorem to get some $c \in (a, b)$ such that $g'(c) = 0$. But we have $g'(x) = f'(x) - h'(x)$ so $f'(c) = h'(c)$. But $h'(c) = m = \frac{f(b)-f(a)}{b-a}$. Hence, we found $c$ such that $f'(c) = \frac{f(b)-f(a)}{b-a}$. □

**Theorem 1.5** (Cauchy MVT)**.** Let $f, g$ be defensible on $(a, b)$ and continuous on $[a, b]$. Then if $g(b) \neq g(a)$ there exists $c \in (a, b)$ such that,

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$$

**Note.** This is a more generalized version of MVT. This implies MVT if $g$ is the trivial function $g(x) = x$.

**Proof.** We attempt to construct a function $h$ again to simplify the problem and apply Rolles theorem. First note that what we want is,

$$(f(b) - f(a))g'(c) - (g(b) - g(a))f'(c) = 0$$

Now Rolles theorem gives us that if $h(a) = h(b) = 0$ then for some $c$ we have $h'(c) = 0$. So in some sense we want $h'(c)$ to be of the form above. Using this intuition construct $h$ as,

$$h(x) = (f(b) - f(a))g(x) - (g(b) - g(a))f(x)$$

Note we get $h(a) = f(b)g(a) - f(a)g(a) - g(b)f(a) - g(a)f(a) = f(b)g(a) - f(a)g(a)$. Similarly $h(b) = f(b)g(b) - f(a)g(b) - g(b)f(b) + g(a)f(b) = g(a)f(b) - f(a)g(b)$. So we have $h(a) = h(b)$. And using rolle we have some $c \in (a, b)$ such that $h'(c) = 0$ or that,

$$h'(c) = g'(c)(f(b) - f(a)) - f'(c)(g(b) - g(a)) = 0$$

which gives us our required result. □

### 1.2.1 Applications

First note that MVT is basically just the first order Taylor series. We can see this as follows. Take $a = x$ and $b = x + h$ then we have

$$\begin{aligned} \frac{f(x+h) - f(x)}{h} &= f'(c) \quad \text{for some } c \text{ between } x \text{ and } x + h \\ f(x+h) &= f(x) + hf'(c) \end{aligned}$$

So the first order taylor approximation would be of form $f(x+h) \approx f(x) + hf'(x)$ for small enough $h$. Now we can further apply MVT to get a second order approximation. First note if we apply MVT to $f'$ we have,

$$\frac{f'(x+h) - f'(x)}{h} = f''(c)$$

Now consider this as a function of $h$ and we have $F'(h) = f'(x+h) - f'(x)$ which gives us $F(h) = f(x+h) - f'(x)h$. So we get,

$$\frac{F'(h)}{h} = f''(c)$$

Now apply Cauchy MVT on the functions $F(h)$ and $h$ between 0 and $h$ to get,

$$\frac{F(h) - F(0)}{\frac{h^2}{2} - 0} = \frac{F'(h)}{h} = f''(c)$$

So we have,

$$\begin{aligned} f(a+h) - f'(a)h - f(a) &= f''(c)\frac{h^2}{2} \\ f(a+h) &= f(a) + hf'(a) + \frac{h^2}{2}f''(c) \end{aligned}$$

Which is the final form of our second order taylor expansion.

**Note.** Note that MVT fails for vectors valued functions from $R^n \to R^m$ where $m > 1$. This is because in the range-space between any two points depending on what components we consider, the slopes might be different which result in varying results. Note that regardless of $n$ if $m = 1$ it still holds as we can parametrize a line in $R^n$ and it becomes $R \to R$ again.

## L'Hopital's rule

**Theorem 1.6** (Version 1). Assume $g$ and $g'$ are nonzero near a and that $\lim_{x\to a} f(x) = 0 = \lim_{x\to a} g(x)$. Note that $f(a), g(a)$ might not be defined. Assuming that $\lim_{x\to a} \frac{f'(x)}{g'(x)}$ exists, then,

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

**Proof.** The only tool till now to connect two functions and their derivative is to use Cauchy MVT which tell us that,

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$$

Now define $\lim_{x\to a} \frac{f'(x)}{g'(x)} = L$ and by definition we have $\forall \varepsilon > 0, \exists \delta$ such that if $x$ is in a delta neighborhood of $a$ then we have,

$$\left| \frac{f'(x)}{g'(x)} - L \right| < \frac{\varepsilon}{100}$$

Now note we established that $\lim_{x\to a} f(x) = 0$ and we have the same for $g(x)$, so in other words for any $\varepsilon$ if $x$ is in some delta neighborhood of $a$ then $f(x)$ and $g(x)$ is in an $\varepsilon$ neighborhood of 0. Choose the smallest $\delta$ now. And now given the delta neighborhood of $a$ i.e $(a - \delta, a + \delta)$ we have,

$$\lim_{x\to a} f(x) = 0$$
$$\lim_{x\to a} g(x) = 0$$

Now pick a $y$ in our $\delta$ ball around $a$ and note that we then have $|f(y) - 0| < \varepsilon$, doing the same for $g$ and using our limit theorems we can now say that,

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim x \to a \frac{f(x) - f(y)}{g(x) - g(y)}$$

for simplicity we can choose $a < y < x$. But note that now the RHS can be written as,

$$\lim_{x\to a} \frac{f(x) - f(y)}{g(x) - g(y)} = \lim_{x\to a} \frac{f'(c)}{g'(c)}$$

where $c$ is between $y$ and $x$ and thus in the $\delta-$neighborhoods given an $\varepsilon$ and hence we have that,

$$\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$$

□

## Clairant's theorem

**Theorem 1.7.** Let $f : U \to \mathbb{R}$ where $U \subseteq R^n$ open. Let $f$ be $C^2$ on $U$, ($\partial_i f, \partial_i\partial_j f$ are continuous $\forall i, j$). Then,

$$\partial_i\partial_j f = \partial_j\partial_i f \text{ for any } i, j \text{ on } u$$

**Proof.** First consider $\partial_i\partial_j f(x, y)$. Now apply MVT to the first coordinate and we have,

$$\frac{\partial_j f(x + h_1, y) - \partial_j f(x, y)}{h_1} = \partial_i\partial_j f(c_1, y)$$

Now define $g(y) = f(x+h_1, y)-f(x, y)$ which gives us $g(y+h_2) = f(x+h_1, y+h_2)-f(x, y+h_2)$ Now using MVT on $g$ we have,

$$\begin{aligned}\frac{g(y + h_2) - g(y)}{h_2} &= \partial_j g(c_2)\\ &= \partial_j f(x + h_1, c_2) - \partial_j f(x, c_2)\\ &= h_1\partial_i\partial_j f(c_1, c_2)\end{aligned}$$

Simplifying the left side we have,

$$\partial_i\partial_j f(c_1, c_2) = \frac{1}{h_1h_2}(f(x + h_1, y + h_2) - f(x, y + h_2) - f(x + h_1, y) + f(x, y))$$

Note here that taking $h$ to zero we are able to choose the value for $c$ given $x, y$ (and we've shown this is true for any $x, y$).

Now in the above equation we see that it's symmetric between both the coordinates. Which means we're able to do the same to $\partial_j\partial_i f(x, y)$ as well to get the same representation and hence proving it's equal. Or we could reverse MVT to get our desired solution. □

## 1.3 Jacobian

**Definition.** Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be (Frechet) differentialble at $p$. Then,

$$Df(p) = f'(p)$$

is a matrix such that,

$$Df(p) \cdot e_j = \partial_j f(p) = (\partial_j f_1(p), \dots, \partial_j f_n(p))$$

**Remark.** $f$ can be written as $f = (f_1, \dots, f_n)$. So this would mean that the $j'th$ column of the Jacobian would be the partial of the $j'th$ component for $f_1, \dots, f_n$

**Note.** We have,

$$\begin{pmatrix}\partial_1 f_1(p) & \dots & \partial_n f_1(p)\\ \vdots & \ddots & \vdots\\ \partial_1 f_n(p) & \dots & \partial_n f_n(p)\end{pmatrix}$$

### Chain rule

Let $f : \mathbb{R}^n \to \mathbb{R}^m$ and $g : R^m \to R^k$ where $f$ is diff (Frechet) at $p$ and $g$ is at $f(p)$. Then we have,

$$(g \circ f)'(p) = g'(f(p))f'(p)$$

So a concrete formula would be,

$$\begin{aligned}
(g \circ f)'(p) &= g'(f(p))f'(p) \\
\partial_1 (g \circ f)(p) &= (g \circ f)'(p) \cdot e_1 \\
&= g'(f(p))f'(p) \cdot e_1 \\
&= g'(f(p))_1 f(p)
\end{aligned}$$

Note here that $g'(f(p))$ is a matrix and $\partial_1 f(p)$ would be the partial wrt the first variable component for each component in $f$. So we have,

$$\begin{aligned}
\partial_1 (g \circ f)(p) &= g'(f(p))_1 f(p) \\
&= g'(f(p)) \left( \sum_{\alpha=1}^{m} \partial_1 f_\alpha(p) e_\alpha \right) \\
&= \sum_{\alpha=1}^{m} \partial_1 f_\alpha(p) g'(f(p)) e_\alpha \\
&= \sum_{\alpha=1}^{m} \partial_\alpha g(f(p)) \partial_1 f_\alpha(p)
\end{aligned}$$

# Chapter 2

# Integration

## 2.1 Riemann integral

Let $f : [a, b] \to \mathbb{R}$ and $f$ be a bounded function. Then for any partition $P$ of $[a, b]$ in the form,

$$P = (a, t_1, t_2 \ldots, t_n, b)$$

where $a < t_1 < \ldots, t_n < b$
we define the mesh of $P$ as,

$$\text{mesh}(P) = \sup_i |t_{i+1} - t_i|$$

**Remark.** So a small mesh means a fine partition and a big mesh means a coarse partition.

**Definition** (Darbaux sum)**.** The Upper Darbaux sum is defined as,

$$U(P, f) = \sum_{i=1}^{n} (t_{i+1} - t_i) \sup_{[t_i, t_{i+1}]} f$$

the Lower Darboux sum is,

$$L(P, f) = \sum_{i=1}^{n} (t_{i+1} - t_i) \inf_{[t_i, t_{i+1}]} f$$

and the Riemann sum is when we pick an arbitrary value $c_i$

**Note.** Note that we have,

$$L(P, f) \leq R(P, f) \leq U(P, f)$$

We say that $f$ is Riemann-integrable when,

$$\lim_{\text{mesh} P \to 0} U(P, f) = \lim_{\text{mesh} P \to 0} L(P, f) = L$$

which also by squeeze theorem implies that $\lim R(P, f) = L$

**Theorem 2.1.** If $f$ is continuous and bounded then $f$ is Riemann integrable.

**Proof.** First note that if $f$ is continuous in a bounded interval (a compact set) then it is uniformly continuous. So fix an $\varepsilon > 0$ and we have a $\delta_\varepsilon$ such that,

$$d_\varepsilon(x, y) < \delta \Rightarrow d(f(x), f(y)) < \varepsilon \quad \forall x, y$$

Note that $d(x, y) < \delta_\varepsilon$ implies that if we have $\text{mesh}(P) < \delta_\varepsilon$ then,

$$|\sup_{|t_{i+1}-t_i|} f - \inf_{|t_{i+1}-t_i|} f| \leq \varepsilon$$

Then we have,

$$\begin{aligned} |U(P, f) - L(P, f)| &\leq \sum_{i=0} \varepsilon |t_{i+1} - t_i| \\ &= \varepsilon \sum_{i=0} |t_{i+1} - t_i| = \varepsilon(b - a) \end{aligned}$$

Note that our partition bounds are fixed i.e $a$ and $b$. So if choose $\delta$ for $\frac{\varepsilon}{|b-a|}$ then we get our desired inequality.

Now, we've shown that $\lim |U - L| \to 0$ which is not to say that the limits $\lim U$ and $\lim L$ exists. In order to prove the existence, we need to show that two arbitrary sequences of $U$ where the mesh of the partition goes to zero converge to the same value.

Let $P$ and $P'$ be two partitions, we define the common refinement of the both as the partition formed by interleaving both the partitions. Let the common refinement be denoted as $P \cup P'$. Now if we have $\text{mesh}(P)$ and $\text{mesh}(P') < \delta$ then we also have $\text{mesh}(P \cup P') < \delta$ and we get,

$$\begin{aligned} U(P \cup P', f) &\leq U(P, f) \\ U(P \cup P', f) &\leq U(P', f) \end{aligned}$$

Now using the uniformly continuity as above we can show that,

$$|U(P \cup P', f) - U(P, f)| \leq \varepsilon$$

and similarly

$$|U(P \cup P', f) - U(P', f)| \leq \varepsilon$$

and we can easily conclude that,

$$|U(P, f) - U(P', f)| \leq \varepsilon$$

Now this implies that two sequences of upper darboux sum wrt to tow sequences of partitions as the mesh of both go to zero converge to the same which imply that there is only one subsequence limit.

□

**Remark.** The reason why we need the common refinement of the partitions are because partitions are partially ordered, i.e we cannot compare every pair of partitions - we can only compare them if one is a subset of the other. However, with the common refinement we can compare two partitions to the common refinement and hence have some common comparison.

## 2.2 Discontinuities

We showed above that a continuous function $f$ on a bounded interval is Riemann Integrable, however, consider now a function $f$ with a discontinuity at point $c$.

**Theorem 2.2.** If $f$ is continuous on $[a, b]$ except at a single point $c$, then $f$is Riemann integrable

**Proof.** $c$ is some point in $[a, b]$. As $f$ has a discontinuity at $c$ that means the uniform continuity breaks when we get close to $c$. So all we need to do is restrict the bad interval around $c$. So consider an interval around $c$, $[c-k, c+k]$, in this interval the maximum height of the function is at most $\sup_f - \inf_f = M$. And now consider our partition as,

$$P = (a, \ldots, c-k, c+k, \ldots, b)$$

where $k \min\{\frac{\varepsilon}{2M}, \delta_\varepsilon\}$, here $\delta$ comes from the delta of the uniform continuity. Now we get,

$$\begin{aligned}
|U(P,f) - L(P,f)| &\leq \frac{\varepsilon}{|b-a|}|c-k-a| + M \cdot 2k + \frac{\varepsilon}{|b-a|}|b-c-k| \\
&\leq \varepsilon + M \cdot 2\frac{\varepsilon}{2M} \\
&= 2\varepsilon
\end{aligned}$$

□

**Note.** Doing the same process we can Riemann integrate any $f$ that has finitely many discontinuities in a bounded set. More specifically where the discontinuities are measure-zero.

More generally now, discontinuities can mainly take form in two ways, a set with Jordan content zero and a set with Lebesgue Measure zero.

**Definition** (Jordan content zero)**.** A set $A$ has Jordan content zero if for any $\varepsilon$ we can find finitely many balls,

$$B_1, \ldots, B_m \text{ such that}$$

$$\sum_{i=1}^{m} \text{length}(B_i) < \varepsilon$$

**Definition** (Lebesgue measure zero)**.** A set $A$ has Lebesgue measure zero if for any $\varepsilon$ we can find a countable set of balls,

$$B_1, B_2, \ldots, \text{ such that}$$

$$\sum_{i=1} \text{length}(B_i) < \varepsilon$$

**Note.** Note that Jordan content zero is stronger as it implies Lebesgue measure zero. If we have a finite number of balls then we have a countable number of balls and hence it is also measure zero

**Note.** It's easy to show that functions with Jordan content zero discontinuity sets that are continuous in a bounded set by considering the finite balls that center the discontinues points and performing the same idea shown in the previous section.

**Example.** The set $\{1\}$ is both Jordan content zero and Lebesgue measure zero. But the set $\mathbb{Q} \cap [0, 1]$ is NOT Jordan content zero (as we cannot cover it with a finite number of balls) but is Lebesgue Measure zero. ◇

**Definition.** The oscillation of a function $f$ is defined as,

$$\text{osc}_f(x) = \inf \text{diameter} f(B(x, r))$$

where the diameter is maximum distance between two points in a set i.e diam $= \sup_{x,y \in E} d(x, y)$

**Note.** Note that as we shrink the radius, the diameter of $f$ decreaes (as the smaller is contained within the larger and hence can only be at most as much as the larger)

**Remark.** If a function is continuous then $osc_f(x) = 0$ and similarly if we have $osc_f(x) = 0$ at all points $x$ then $f$ is continuous (as for any $\varepsilon$ as we have oscillations as zero we can find a radius - the $\delta$ - small enough that the points inside are arbitrarily close and hence by definition we have continuity)

**Theorem 2.3.** If a function $f$ is continuous on $[a, b]$ with a measure zero discontinuity set, then $f$ is Riemann integrable.

**Proof.** As $f$ is continuous we have that the $osc_f(x) = 0$ where $x$ is not discontinuous is zero. And we can say that,

$$Dis(f) = \{x : osc(x) > 0\}$$

Now define two sets,

$$D_n = \left\{x : osc_x \geq \frac{1}{n}\right\}$$
$$D_n^c = \left\{x : osc_x < \frac{1}{n}\right\}$$

Note that $D_n$ is closed and $D_n^c$ is open. Now we can define,

$$Dis(f) = \cup_{n=1}^{\infty} D_n$$

So we have $Dis(f)$ is the countable union of closed sets. Now note that because $Dis(f)$ is also measure zero we can cover it with countably many balls. Clearly we have $D_n$ is also covered by countably many balls. But note $D_n$ is compact and hence we can find a finite subcover of these sets to cover it. More specifically we can find $N = N(n)$ such that,

$$D_n \subseteq \bigcup_{i=1}^{N} B_i$$

so that $B_1, \ldots, B_N$ gives finite bad intervals that we need to restrict width for. Now on the other hand for,

$$I \subseteq D_n^c$$

we have $U - L$ (upper - lower) as,

$$|U - L| \leq \frac{1}{n}\text{diam}(I) \leq \frac{\varepsilon}{10|b-a|}\text{diam}(I)$$

Now note that summing over all intervals $I$ will still be only as big as $|b - a|$ which means we still have $\frac{\varepsilon}{10}$ to work with for the finite bad intervals we want to restrict. $\square$

**Theorem 2.4** (Weak Dominated Convergence). Let $f \geq 0$ be Riemann integrable on $[a, b]$ with $f_n(x) \to f(x)$ as $n \to \infty$ for all $x \in [a, b]$ (except for a possibly measure-zero subset), then,

$$\int_a^b f_n \to \int_a^b f$$

as $n \to \infty$

we can use this to interchange the order to derivative and integral as follows,

$$\begin{aligned}\partial_x\left(\int_a^b f(x,y)dy\right) &= \lim_{h\to 0}\int_a^b \frac{f(x+h,y)-f(x,y)}{h}dy \\ &= \int_a^b \lim_{h\to 0}\frac{f(x+h,y)-f(x,y)}{h}dy \\ &= \int_a^b \partial f(x,y)dy\end{aligned}$$

Now if,

$$g_{n,x}(y) = \frac{f(x+h,y)-f(x,y)}{h}$$

Note that we can only apply this when each $d_{n,x}(y)$ is dominated by some function that is a Riemann Integrable function. So if, $f$ is $C^1$ by MVT we have,

$$g_{n,x}(y) = \partial_1 f(c_{n,x},y)$$

where $c_n$ is between $x$ and $x+h$ and as $\partial f$ is bounded and continuous by $C^1$ so is $g$ and hence we can use the theorem.

## 2.3 Fundamental Theorem of Calculus

**Theorem 2.5.** Let $f$ be a continuous function on $[a,b]$ such that $\forall x \in [a,b]$ we have,

$$\partial_x\left(\int_a^x f(x,y)d(y)\right) = f(x)$$

**Proof.** We have,

$$\partial_x\left(\int_a^x f(x,y)d(y)\right) = \lim_{h\to 0}\frac{1}{h}\left(\int_a^{x+h} f - \int_a^x f\right)$$

For $\varepsilon > 0$ as $f$ is continuous we have $\delta > 0$ such that for all $|h| < h_0$ we have $|f(x+h)-f(x)| < \varepsilon$. Now note that $\left(\int_x^{x+h} f\right) - f(x) = \int_x^{x+h} f - f(x)$ and hence we have,

$$\left|\int_x^{x+h} f - f(x)\right| \leq \varepsilon$$

□

**Theorem 2.6.** If $f \in C^1$ on $[a,b]$ then,

$$\int_a^b f' = f(b) - f(a)$$

**Proof.** Note we can write $\int_a^x f' = f(x) - f(a)$. Now using the theorem from above we have,

$$\partial_x\left(\int_a^x f'\right) = f'(x)$$

and note that we also have $\partial_x(f(x)-f(a)) = f'(x)$ which means that the function only differs by a constant. That is because we have,

$$\partial_x(LHS - RHS) = 0$$

We have that $LHS - RHS = \textit{constant}$ and as $LHS(a) = 0 = RHS(a)$ we have the constant as zero and hence $LHS = RHS$ $\square$

## 2.4 Integral in $\mathbb{R}^n$

We have $U \subseteq R^n$ a bounded open set i.e $\overline{U} = U \cup \partial U$. Assume that $\partial U$ is piecewise continuous. Here we partition $U$ into squares with disjoint interior. The mesh of these squares are defined as the diameter. So our partition is $P = \bigcup_i s_i$ where $s_i$ is a square. Then we have mesh $P = \max_i \text{diam} s_i$. We can then define the upper, lower and riemann sum normally,

$$U(P, f) = \sum_i \left(\sup_{s_i} f\right) \text{vol}(s_i)$$

we can define the lower and riemann integral similarly.

### 2.4.1 Indicator function

Let $E \subseteq R^n$ then,

$$1_E(x) = \begin{cases} 1 & \text{if } x \in E \\ 0 & \text{if } x \notin E \end{cases}$$

### 2.4.2 Simple function

Linear combination of indicators.

**Example.** Let $f$ be bounded on $U$ as above and define,

$$F = \sum_i 1_{s_i} f(\text{center of } s_i)$$

note that here $F$ is basically the discretized version of $f$, in that for a given "cube" it has the same value which is just a scaled version of the indicator. $\diamond$

Here if we have,

$$\int_U F = \sum_i \text{vol}(s_i) f(\text{center of } s_i) = R(P, f)$$

Now note that as we take the mesh of the partitions to zero i.e the cubes become smaller and smaller, we end up approximating the true function $f$. Hence, the limit of a simple function as the partition goes to zero ends up being our underlying function.

In other words if $s_n$ is our simple function such that $\lim_{n\to\infty} s_n(x) = f(x)$ then using DCT we have,

$$\lim_{n\to\infty} \int_U s_n = \int_U \lim_{n\to\infty} s_n = \int_U f$$

But note the first is just the limit of rieman sum which is the definition of the riemann integral.

We can also extend this to function $f$ that have measure-zero discontinuity, where in $R^n$ any set that are "k-dim" objects where $k < n$ are measure zero sets. For instance $[0, 1]$ has measure zero in $R^2$.

## 2.5 Change of Variables

Let $f$ be $C^1$ on $\overline{U}$ and let $v \subseteq R^n$ be open and bounded. And let,

$$\phi : \overline{V} \to \overline{U} \text{ be bijective and } C^1$$

and $\forall y \in \overline{V} : \phi'(y)$ is invertible, then,

$$\int_{\overline{U}} f = \int_{V} |\det \phi'|(f \circ \phi)$$

**Proof.** In order to prove this, it's enough to show it's true for indicator functions. So let, $f = 1_E$, $f \circ \phi = 1_{\phi^{-1}(E)}$ then we have,

$$\int f = \text{vol } E$$

And we have $\int (f \circ \phi)|\det \phi'| = \int_{\phi^{-1}(E)} |\det \phi'|$ □

## 2.6 Extra topics on integration

1. **Integration is linear:** We have,

$$\int_a^b (\lambda_1 f + \lambda_2 g) = \lambda_1 \int_a^b f + \lambda_2 \int_a^b g$$

   This is straightforward when we think of integrals as the limit of the riemann sums.

2. **Triangle Inequality:** We traditionally have $|a + b| \leq |a| + |b|$, extending this we get,

$$\left| \int_a^b f \right| \leq \int_a^b |f|$$

3. If $f$ is riemannn-integrable on $[a, b]$ and if we have,

$$\int_a^b |f| = 0$$

   then $f = 0$ on $Dis(f)^c$

4. **Holder Inequality:** We have,

$$\int_a^b |fg| \leq \left( \int_a^b |f|^p \right)^{1/p} \left( \int_a^b |g|^q \right)^{1/q}$$

   where $\frac{1}{p} + \frac{1}{q} = 1$. Note here if $q \to \infty$ then we have,

$$\left( \int_a^b |g|^q \right)^{1/q} \to \max_{[a,b]} |g|$$

   In this context the notation $||f||_{L^p}$ is equal to $\left( \int |f(x)|^p \right)^{1/p}$

5. **Integration by Parts:** Is simply FTC on a function $h = fg$ we have,

$$h(b) - h(a) = \int_a^b h'(x)\ dx$$
$$f(b)g(b) - f(a)g(a) = \int_a^b f'g + g'f\ dx$$
$$\int_a^b f'g = [f(y)g(y)]_a^b - \int_a^b g'f$$

Note that this is essentially integration by parts if we consider $f'$

6. **Improper Integral:** To show improper integrals the goal is for a set $U$ we need to show that any sequence of sets that are contained in each other ie. $E_1, E_2, \ldots$ where $E_i \subseteq E_{i+1}$ where $E_i \to U$ will converge to the same integral.

   Note that we can only do this for $f$ that is increasing for instance alternating signs series can cancel out to different values based on how you do it. Now if we know that $f$ is increasing then we can use a common refinement of two arbitrary sequence $E_i \to U$ and $F_i \to U$. This gives us the same integral for two arbitrary sequence and hence establishes the improper integral.

### Hilbert Space

A Hilbert space is a vector space that has an inner product and is complete (Cauchy sequences converge). So if we consider the $L^2$ space, this is a Hilbert space as we have a natural inner product,

$$\langle f, g\rangle_{L^2} = \int f\overline{g}$$

If we consider the trace of the jacobian i.e the divergence $\Delta f(x, y) = (\partial_{11} + \partial_{22})f$, the spectral theorem for $\Delta$ on $L^2$ is the Fourier transform, this gives rise to harmonic analysis.

## 2.7 Fubini

**Theorem 2.7.** Let $f \in L^1([a, b] \times [c, d])$ then,

$$\int_c^d \int_a^b f(x, y)\ dx\ dy = \int_a^b \int_d^b f(x, y)\ dy\ dx$$

**Proof.** First we split $f$ into real and imaginary parts and WLOG assume $f$ is real valued. Now split $f$ into the positive and negative parts so we have $f = f^+ - f^-$, so WLOG assume that $f \geq 0$. Now note that it is enough to show for indicator functions (of cubes) as then we have simple function of cubes which using DCT gives us integrable functions on $R^2$.

So let $f = 1_{[a_0,b_0]\times[c_0,d_0]}$ now the problem is essentially to show that $(d_0 - c_0)(b_0 - a_0) = (b_0 - a_0)(d_0 - c_0)$ which is obviously true. $\square$

## 2.8 High-dimensional FTC

Let $f$ be $C^2$ on $[a,b] \times [c,d]$. Then we have,

$$\begin{aligned}
\int_c^d \int_a^b \partial_1 \partial_2 f(x,y)\ dx\ dy &= \int_c^d (\partial_2 f(b,y) - \partial_2 f(a,y))\ dy \\
&= \int_c^d \partial_2 f(b,y) - \int_c^d \partial_2 f(a,y)\ dy \\
&= f(b,d) - f(b,c) - (f(a,d) - f(a,c)) \\
&= f(b,d) - f(b,c) - f(a,d) + f(a,c)
\end{aligned}$$

## 2.9 Green's Theorem

It is enough to show greens theorem for an arbitrary rectangle as we can then generalize this by splitting any set into a bunch of rectangles. So let $E = [a,b] \times [c,d] \subseteq R^2$ then if $f = (f_1, f_2)$ then we have curl $f = \partial_1 f_2 - \partial_2 f_1$ and we get,

$$\begin{aligned}
\int_c^d \int_a^b (\partial_1 f_2 - \partial_2 f_1)\ dx\ dy &= \int_c^d \int_a^b \partial_1 f_2\ dx\ dy - \int_c^d \int_a^b \partial_2 f_1\ dx\ dy \\
&= \int_c^d \int_a^b \partial_1 f_2\ dx\ dy - \int_a^b \int_c^d \partial_2 f_1\ dy\ dx \\
&= \int_c^d f_2(a,y) - f_2(b,y)\ dy - \int_a^b f_1(x,d) - f_1(x,c) dx \\
&= \int_c^d f_2(a,y) - \int_c^d f_2(b,y) - \int_a^b f_1(x,d) + \int_a^b f_1(x,c)
\end{aligned}$$

But note the quantity on the right is the integral over the boundary for the rectangle

## 2.10 Divergence Theorem

Divergence theorem is shown in HW5, the idea is to use partition of unity to get our ball as the union of open balls, and then we write $f = \sum f_i$ where each $f_i$ has compact support. Then we use spherical coordinates.

## 2.11 Stokes Theorem

Stokes theorem is shown in Problem 3, HW5. We can use the exterior derivative.

## 2.12 Inverse Function Theorem

Shown in Problem 2 of HW5. The idea of inverse function theorem is that if $f$ is $C^1$ and our derivative is invertible at a point (so the determinant of the Jacobian is non-zero). Then for a neighborhood around that point $f$ has an inverse which is also $C^1$.

## 2.13 Implicit Function Theorem

We can use the above notion of an inverse function to get the implicit function theorem.

**Theorem 2.8.** Let $f : U \to R$ where $U \subseteq R^2$ an open set and $f$ is $C^1$. Now let $(x_0, y_0) \in U$ be such that,

$$\partial_y f(x_0, y_0) \neq 0$$

and let $a = f(x_0, y_0)$ then locally near $(x_0, y_0)$ the level set $\{f = a\}$ is a graph i.e $y = g(x)$

**Note.** The condition $\partial_y f(x_0, y_0) \neq 0$ means the curve is not vertical. Think the unit circle in $R^2$, in this case the left and right endpoint have $\partial_y$ as zero i.e a vertical line. The point of this theorem is that for all the other points we can essentially a neighborhood around that point where we can flatten it.

*Proof.* Define $H :_0\to V_0$ as $(x, y) \to (x, f(x, y))$. Note that for the level set $\{f = a\}$ this just maps to $(x, a)$ i.e a straight horizontal line. Now we can easily show that $\det H'(x_0, y_0)e_0$ as it will just be equal to $\partial_y f(x_0, y_0)$ which we know is non-zero. So use IFT for $H$ and there is an inverse function $H^{-1}$ at that neighborhood that maps back to $(x, y)$. But now let $\pi_2$ be the function that projects onto $y$ axis. Then we have ,

$$y = \pi_2(H^{-1}(x, a))$$

So essentially for the level curve $\{f = a\}$ we found a function $g(x) = \pi_2(H^{-1}(x, a)$ such that $y = g(x)$ $\square$

**Remark.** We have $f(x, g(x)) = a$ for $x$ near $x_0$ if we differentiate it we get,

$$\partial_x f(x, g(x)) + \partial_y f(x, g(x))g'(x) = 0$$

this is essentially,

$$\frac{\partial y}{\partial x} = g'(x) = -\frac{\partial_x f}{\partial_y f}$$

The implicit function theorem essentially tells us that regular level sets of smooth function are manifolds.

## 2.14 Generalization of IFT

We can generalize IFT to high dimentions i.e $x \in R^m$ and $y \in R^n$. So consider we have a functoin,

$$f : R^m \times R^n \to R^n, C^1$$

Similarly say $\partial_y f(x_0, y_0)$ is invertible and if $a = f(x_0, y_0)$ then $\{f = a\}$ is locally a graph i.e $y = g(x)$

Let $f : U \subseteq R^{m+n} \to R^n$. Let $p \in U$ and if $f'(p)$ is surjective then locally we can straighten to turn $f$ into $(x, y) \to y$. That is, we can write $y = g(x)$ where $x \in R^m$ and $y \in R^n$

*Proof.* Let $A = f'(p)$ i.e the Jacobian. Note that $A$ is a $n \times (m + n)$ matrix. Now if $A$ is surjective we know that the columns span all of $R^n$. So we can find a basis of $R^n$ in the columns. Take this collection of vectors ($n$ vectors) to be $y$ and the rest $m$ vectors to be for $x$. Now we have $\partial_y f(x_0, y_0)$ is invertible. $\square$

## 2.15 Constant rank theorem

**Theorem 2.9.** Let $f : U \subseteq R^{m+n}$ be $C^1$ and $U$ be $R^n$. If $p_0 \in U$ and $f'(p)$ has constant rank $r$ for $p$ near $p_0$. Then we can locally straighten $f$ into ,

$$(x, y) \to (y_1, \ldots, y_r, 0, \ldots, 0)$$

**Proof.** Proof is in HW6 P1 $\square$

## 2.16 Lagrange multipliers

Let $G : \mathbb{R}^n \to \mathbb{R}$ and $g : U \subseteq \mathbb{R}^n \to \mathbb{R}$. Note that we have $M = \{G = a\}$ a compact constraint. We assume that $G'$ is surjective on $M$. Now let $f$ be $C^1$ on a neighborhood of $M$. We need to optimize $f$ on $M$.

First assume that $p_0 = (\overline{p}, p^*)$ where $\overline{p} \in \mathbb{R}^{n-1}$ and $p \in \mathbb{R}$. Now by IFT, WLOG we have locally $M$ (near $p_0$) is a graph and $\partial_{x_n} G(p_0) \neq 0$. Now because it's a graph we have $x_n = g(x_1, \dots, x_{n-1}) = g(\overline{x})$. Now for each $i = 1, \dots, n-1$ we have,

$$\partial_{x_i} \left( f(\overline{x}, g(\overline{x})) \right) = 0$$

so this means that,

$$\begin{aligned} \partial_{x_i} f(\overline{p}, g(\overline{p})) + \partial_{x_n} f(\overline{p}, g(\overline{p})) \partial_{x_i} g(\overline{p}) &= 0 \\ \partial_{x_i} f(\overline{p}, g(\overline{p})) &= -\partial_{x_n} f(\overline{p}, g(\overline{p})) \partial_{x_i} g(\overline{p}) \end{aligned}$$

Now observe that $G(\overline{x}, g(\overline{x})) = a$ where $a$ is the max / min. So we have,

$$\begin{aligned} \partial_{x_i} (G(\overline{x}, g(\overline{x}))) &= 0 \\ \partial_{x_i} G(\overline{p}, g(\overline{p})) = -\partial_{x_n} G(\overline{p}, g(\overline{p})) \partial_{x_i} g(\overline{p}) \end{aligned}$$

Now dividing the two we have,

$$\nabla f(p_0) = \lambda \nabla G(p_0)$$

where

$$\lambda = \frac{\partial_{x_n} f}{\partial_{x_n} G}(p_0)$$

## 2.17 Sard's theorem

**Theorem 2.10.** Let $f : U \to R^n$ with $U \subseteq R^n$ open and $f$ is smooth. Now first any $y \in R^n$ is called a regular value when $f'$ is surjective on $\{f = y\}$. **Sards Theorem** says that the set of critical values (non-regular values) has measure zero (in $R^n$).

**Note.** This theorem essentially states that most level sets are manifolds

**Example.** Consider $f : \mathbb{R}^2 \to \mathbb{R}$ with $f(x, y) = x^2 + y^2$. Now take the level set $\{x^2 + y^2 = a\}$. Here for $a \neq 0$ we have regular values but $a = 0$ is a critical value and note that this is a measure zero set (finite amount of elements in this case). $\diamond$

**Proof.** The proof is due to consatnt rank theorem. That is, where $f'$ has local ran $r < n$ then $f$ maps locally to a submanifold of dim $< n$ which is measure zero in $\mathbb{R}^n$. $\square$

# Chapter 3

# Additional Topics

## 3.1 Contraction Mapping Theorem

**Theorem 3.1.** Let $E$ be a Cuachy-complete metric space (meaning every Cauchy sequence converges within the set) and let $f : E \to E$ be continuous. Let $\alpha \in (0, 1)$ and if we have,

$$|f(x) - f(y)| \leq \alpha|x - y|, \quad \forall x, y \in E$$

then there is a unique fixed point $a \in E$ such that,

$$f(a) = a$$

**Proof.** Idea is to consider the sequence $x, f(x), f(f(x)), \dots$ and show that this is a Cauchy sequence that converges to a fixed point. Full proof in HW6, Q2 $\square$

## 3.2 Picards's Theorem (for autonomous ODE)

**Theorem 3.2.** Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be smooth, then the ODE,

$$x'(t) = F(x(t))$$
$$x(0) = p \in \mathbb{R}^n$$

has a local-in-time solution i.e $x : (-\varepsilon, \varepsilon) \to \mathbb{R}^n$.

**Proof.** We essentially need to solve (using FTC),

$$x(t) - x(0) = \int_0^t F(x(s))\ ds$$
$$x(t) = p + \int_0^t F(x(s))\ ds$$

Now define $\Phi$ (an operator so $\Phi(x)$ takes in the function $x$ and outputs another function $\Phi(x)$ a function of $t$) where

$$\Phi(x)(t) = p + \int_0^1 F(x(s))\ ds$$

Now our question becomes,

$$x = \Phi(x)$$

Now note that if we pick our domain as,

$$X = C^0([-\varepsilon, \varepsilon] \to \overline{B(p,R)})$$

Now note that $\overline{B(p,R)}$ is Cauchy complete. Note that we still need to choose $\varepsilon$ and $R$.

First pick $R = 1$. Let $M_R = \max_{B(p,R)}\{|F|, |F'|\}$. Then assuming $x \in X : x(s) \in B(p,R)$ for all $s$. We have,

$$\begin{aligned}\left|\int_0^t F(x(s))\right| &\leq |t| M_R \\ &\leq \varepsilon M_R\end{aligned}$$

Now we want $\varepsilon M_R < R$. So simply pick,

$$\varepsilon = \frac{R}{100 M_R} \text{ (for } R = 1)$$

Now this means we have $\Phi : X \to X$ and for contraction we need,

$$d_x(\Phi(x), \Phi(y)) \leq \alpha d_x(x, y)$$

we see this as follows,

$$\begin{aligned}\sup_t |\Phi(x)(t) - \Phi(y)(t)| &= \sup_t \left|\int_0^t (F(x(s)) - F(y(s)))\right| \\ &\leq \sup_t \left|\int_0^t (\max_{B(p,R)} |F'|)|x(s) - y(s)|\right| \\ &\leq \varepsilon \max |F'| d_x(x, y)\end{aligned}$$

and we have $\max |F'| \leq M_R$ and hence we have a contraction which implies the existence of a solution and local uniqueness.

Now we need to show global uniqueness... □

## 3.3 Foundational ODE

Consider the ODE $f'(t) = f(t)$ and $f(0) = 1$ where $f$ is real valued. Now Piccards theorem gives us a unique solution to this problem, we call this the exponential function.Here we have the following,

1. $f$ is smooth
2. $f'(0) = f(0) = 1$
3. By Taylors theorem we have,

$$\begin{aligned}f(x) &= f(0) + f'(0)x + \frac{f^{(2)}(0)}{2!}x^2 + \ldots \\ &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \\ &= \sum_{k=0}^{\infty} \frac{x^{k+1}}{(k+1)!} = e^x\end{aligned}$$

4. We have $e^x e^y = e^{x+y}$. We see this by defining $f(x) = e^x e^y$ and $g(x) = e^{x+y}$. Now note that $f'(x) = e^x e^y = f(x)$ and $f(0) = e^y$, similarly note that $g'(x) = g(x)$ and $g(0) = e^y$. But by picards the solution to this is the same i.e $f = g$.

5. $f(x) > 0, \forall x \in \mathbb{R}$. Note that for positive $x$ we trivially have $f(x) > 0$ because of the Taylor expansion. Now consider $e^0 = 1$ note that by the previous thing we also have for any $x$ that $e^0 = e^{x-x} = e^x e^{-x}$. Now choose any $x \in \mathbb{R}$ and note that either $x$ or $-x$ is positive. Consider WLOG $e^x$ is positive so we have $e^x e^{-x} = 1$ and $e^x$ is positive which gives us $e^{-x}$ is positive. So clearly for every $x$ we need $e^x$ as positive.

6. $e : \mathbb{R} \to (0, \infty)$ is bijective.

   **Proof.** First note that $e$ is strictly increasing, which makes it injective. Now note that $e^n$ as $n \to \infty$ is $\infty$ and $e^{-n}$ as $n \to \infty$ is 0. And because $e$ is smooth by IVT we have it's surjective. Hence, we showed it's bijective. □

7. Defining $\ln x$. Because it's bijective we define the inverse as $\ln x : (0, \infty) \to \mathbb{R}$. So we have $\exp(\ln x) = x$. Chain rule gives us,

$$\begin{aligned} \exp(\ln x)' &= x' \\ \exp(\ln x)(\ln x)' &= 1 \\ x(\ln x)' &= 1 \\ (\ln x)' &= \frac{1}{x} \end{aligned}$$

## 3.4 Normed Vector Space

Let $V$ be a vector space over $\mathscr{F}$. A norm $\|\cdot\|$ is a function $\|\cdot\| : V \to [0, \infty)$ such that,

1. $\|v\| = 0 \Leftrightarrow v = 0$
2. $\|\lambda v\| = |\lambda| \|v\|, \forall \lambda \in \mathscr{F}, \forall v \in V$
3. $\|u + v\| \leq \|u\| + \|v\|, \forall u, v \in V$

The norm induces a metric space,

$$d(x, y) = \|x - y\|$$

this allows us to define concepts such as open, closed, convergence and Cauchy sequences.

If a Normed Vector Space is Cauchy Complete (meaning every Cauchy sequence converges within the vector space) we call it a Banach space.

**Example.** $R^n$ with Euclidian norm,

$$\|v\| = \sqrt{v_1^2 + \cdots + v_n^2}$$

is a Banach space. ◇

Now let $U, V$ be Banach, define,

$$L(U, V) = \{f : U \to V \mid f \text{ is linear and continuous }\}$$

Now we can define an operator norm on $L$ itself as,

$$\|f\|_{L(U,V)} = \sup_{u \neq 0} \frac{\|f(u)\|_V}{\|u\|_U} = \sup_{\|u\|=1} \frac{\|f(u)\|}{\|u\|}$$

So here $L$ is called the bounded linear map.

**Theorem 3.3.** Now we can show that $f$ is continuous $\Leftrightarrow$ $f$ is bounded.

**Remark.** Note that if $f$ is continuous then for any $x_n \to x \Rightarrow fx_n \to fx$, which is the same as $x_n - x = y_n \to 0 \Rightarrow f(x_n - x) = fy_n \to 0$.

**Proof.** First assume that $f$ is bounded (on the unit ball). So we have,

$$\|y\| \leq 1 \Rightarrow \|fy\| \leq M$$

Note from the remark above that it's enough to show that if $y \to 0$ then $fy \to 0$. So now note for all $y$ we have $\|fy\| \leq M\|y\|$. So trivially we see that $y_n \to 0$ implies that $\|fy_n\| \leq M\|y_n\| \to 0$, hence $f$ is continuous.

Now assume that $f$ is continuous. So fix an $\varepsilon > 0$, then we have a $\delta_\varepsilon > 0$ such that,

$$\|x\| \leq \delta_\varepsilon \Rightarrow \|fx\| \leq \varepsilon$$

so,

$$\|x\| \leq 1 \Rightarrow \|fx\| \leq \delta_\varepsilon^{-1}\varepsilon$$

hence we showed boundedness in the unitball. □

### 3.4.1 Sub-multiplicative

Let $T_1 \in L(U_0, U_2)$ and $T_2 \in L(U_1, U_2)$, then,

$$\|T_2 \circ T_1\| \leq \|T_2\|\|T_1\|$$

**Proof.** We have,

$$\frac{|T_2 T_1 v|}{|v|} \leq \frac{\|T_2\||T_1 v|}{|v|}$$

smaller than $\|T_2\|\|T_1\|$. Now just take sup over it. □

Essentially this gives us that $\|T^n\| \leq \|T\|$.

**Corollary 3.4.** We can show that $\exp(T) = \sum_{k=0}^{\infty} \frac{T^k}{k!} \in L(V, V)$

**Proof.** Proof is HW8. Essentially we just need to show that the infinite sum converges within the set. Do this by considering the partial sums and showing it's Cauchy and because $L(V)$ is Cuachy complete, it converges within the set. □

### 3.4.2 Inner Product

Let $V$ be a complex vector space. The Inner product is a function,

$$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{C}$$

If $V$ is an inner product space that is also Banach, we call $V$ a Hilbert Space. So a hilbert space has,

$$\text{vector space} + \text{topology (norm)} + \text{complete (cauchy)} + \text{geometry (inner product)}$$

### 3.4.3 $L^p$ spaces

For any $f \in C_c^\infty(\mathbb{R}^n)$ (smooth function with compact support) we define the $L^p$ norm as,

$$\|f\|_{L^p} = \left(\int |f|^p\right)^{\frac{1}{p}}$$

Now if we complete $C_c^\infty(\mathbb{R}^n)$ under the $\|\cdot\|_p$ norm we get the $L^p(\mathbb{R}^n)$ spaces. So note that this new space does not necessarily need to have smooth functions, for instance the limit of a series of functions could be the step function which is not smooth.

**Note.** that all $L^p$ spaces are Banach as we define it by the Cauchy-completeness and we also have norms.

When we take $p = 2$ we have our inner product as,

$$\langle f, g\rangle_{L^2} = \int_{\mathbb{R}^n} f\overline{g}$$

so this makes $L^2$ a Hilbert space.

**Remark.** In fact none of the other norms is induced by an inner product for the other $L^p$ spaces, so $L^2$ is the only $L^p$ space that is Hilbert

### 3.4.4 Pythagoras

Let $V$ be Hilbert and let $x, y \in V$, then,

$$\langle x, y\rangle = 0$$

this implies that $\|x+y\|^2 = \|x\|^2 + \|y\|^2 \Rightarrow Re(\langle x, y\rangle = 0$

### 3.4.5 Cauchy-Schwartz

Let $x, y \in V$ with $V$ Hilbert. Then,

$$|\langle x, y\rangle| \leq \|x\|\|y\|$$

**Proof.** We can replace $x$ with $\alpha x$ and now note that for any $t \in \mathbb{R}$ we have,

$$\langle tx - y, tx - y\rangle \geq 0$$
$$\|x^2\|t^2 - 2\langle x, y\rangle t + \|y\|^2 \geq 0$$

this means that,

$$|\langle x, y\rangle|^2 \leq \|x\|^2\|y\|^2$$

□

### 3.4.6 Spectral Theorem

**Theorem 3.5.** Let $A$ be a $n \times n$ matrix which is symmetric. Then fix an orthonormal basis $(b_i)_{i=1}^n$ of $\mathbb{R}^n$ such that,

$$Ab_i = \lambda_i b_i$$

when $\lambda_i \in \mathbb{R}$. Let $(e_i)_{i=1}^n$ be the canonical basis of $\mathbb{R}^n$. Then there is a change of basis matrix $P$ (between $B$ and $E$) such that,

$$A = P \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} P^T$$

where $P^T = P^{-1}$ is an orthogonal matrix.

## 3.5 Torus

The 1-D torus is $\mathbb{T} = \mathbb{R} \mod \mathbb{Z}$, this is the quotient notation, so essentially we're considering the equivalence classes modulo $\mathbb{Z}$. For instance the value $0.3, 1.3, 2.3, -4.7$ all belong to the same equivalence class. Essentially the Torus is $[0, 1)$ in this case where we wrap around to 0 from the right side. We see that this is also homeomorphic to $S^1$ (the circle). More traditionally for the circle consider $\mathbb{R} \mod 2\pi\mathbb{Z}$ i.e $[0, 2\pi)$.

$\mathbb{T}$ is also called a periodic domain and is also compact. We define $\mathbb{T}^n = \mathbb{T} \times \cdots \times \mathbb{T}$.

Consider smooth functions defined on the torus i.e $C^\infty(\mathbb{T}^n)$ then $L^p(\mathbb{T}^n)$ is the Cauchy competition of this (and hence is Banach) under the $L^p$ norm.

$$\|f\|_{L^p} = \left( \int_{\mathbb{T}^n} |f|^p \right)^{\frac{1}{p}}$$

### 3.5.1 Hilbert Spaces

Let $V$ be a Hilbert Spaces (Banach + innerproduct).

1. $B \subseteq V$ is an (algebraic) basis when,
   (a) $B$ is linearly independent
   (b) Span $B = V$
2. $B \subseteq V$ is a Hilbert Basis when
   (a) $B$ is linearly independent
   (b) Closure of Span $B = V$

**Example.** $L^2$ has a countable Hilbert basis but uncountable algebraic basis. Let $\{e_1, e_2, \dots\}$ be an orthonormal Hilbert basis of $V$. ◇

Let $V_n = span\{e_1, \dots, e_n\}$. Then for any $v \in V$ (note not $V_n$, $V$ is the Hilbert space) we have,

$$\text{proj}_{V_n}(v) = \sum_{n=1}^{n} (v \cdot e_i) e_i$$

### 3.5.2 Fourier transform on 1D torus

Let $f \in C(\mathbb{T} \to \mathbb{C})$ so $f$ is in the space of continuous functions from the torus to the complex numbers. Then the fourier transform of $f$ is a function $\hat{f}$ from $\mathbb{Z} \to \mathbb{C}$,

$$\hat{f}(k) = \int_{\mathbb{T}} f(x) e^{-i2\pi kx} \, dx$$

We know that $L^2(\mathbb{T})$ is a Hilbert space (because the $L^2$ norm is induced from an inner product) with Hilbert basis $\{e_k\}_{k\in\mathbb{Z}}$ where,

$$e_k = e^{i2\pi kx}$$

We have,

$$\Delta e_k = e_k'' = -(2\pi)^2|k|^2 e_k$$

So $e_k$ is an eigenvector (eigenfucntion rather) of $\Delta$. We have,

$$\|e_k\|_{L^2} = \left(\int_{\mathbb{T}} |e_k|^2\right)^{\frac{1}{2}} = 1$$

And for all $k \neq l$ we have,

$$e_k \cdot e_l = \int_{\mathbb{T}} e^{ipikx}\overline{e^{i2\pi lx}}\ dx = 0$$

So this gives us,

$$\sum_{k\in\mathbb{Z}} (f \cdot e_k)e_k \to^{L^2} f$$

Note that the left side is essentially projecting $f$ onto each of the basis. Using inner product on $L^2$ we have $f \cdot e_k = \hat{f}(k)$

What this means is that $f \in L^2$ can be decomposed into an infinite sum where each term describes $f$ at a certain frequency i.e $\hat{f}(k)e_k$ is $f$ at frequency $k$.

Another way to see it is that the Fourier transform diagonalzes the $\Delta$ (laplacian) operator and other differential operators (so we're able to turn a PDE problem into a polynomial or ODE problem).

### 3.5.3 Fourier on nD torus

Instead of taking $k \in \mathbb{Z}$ we take $k \in \mathbb{Z}^n$. So $\forall x \in \mathbb{T}^n$ we have $e_k(x) = e^{i2\pi(k\cdot x)}$. Note that $\{e_k\}_{k\in\mathbb{Z}^n}$ is still orthonormal. So we have,

$$\hat{f}(k) = \int_{\mathbb{T}^n} f(x)e^{-i2\pi(k\cdot x)}\ dx$$

which gives us,

$$f =^{L^2} \sum_{k\in Z^n} \hat{f}(k)e_k$$

**Example.** Let $g \in C^\infty(\mathbb{T}^n)$, we need to find $f$ such that $f = g$. ◇

**Solution.** We can apply the Fourier transform to both sides to get,

$$-(2\pi)^2|k|^2\hat{f}(k) = \hat{g}(k)$$

Note that this is only solvable when $\hat{g}(0) = 0 = \int_{\mathbb{T}^n} g$. Now simply take for $k \neq 0$,

$$\hat{f}(k) = -\frac{\hat{g}(k)}{(2\pi)^2|k|^2}$$

where $\hat{f}(0)$ can be any constant. Then we get,

$$f = C + \sum_{k\neq 0} \hat{f}(k)e_k$$

### 3.5.4 Pythagoras

If $u \perp v$ then we have,

$$\|\lambda_1 u + \lambda_2 v\|^2 = |\lambda_1|^2 \|u\|^2 + |\lambda_2|^2 \|v\|^2$$

Similarly we have,

$$\begin{aligned} \|f\|_{L^2}^2 &= \|\sum_{k\in Z^n}^{\infty} \hat{f}(k) e_k\|_{L^2}^2 \\ &= \sum_{k\in Z^n}^{\infty} |\hat{f}(k)|^2 \end{aligned}$$

So this essentially tells us that,

$$\|f\|_{L^2(\mathbb{T}^n)} = \|\hat{f}\|_{L^2(\mathbb{Z}^n)}$$

So essentially the Fourier transform is an invertible isometry (a distance preserving map between metric spaces that preserves, length angles and areas).

Properties of this map are,

1. Linear (so the Fourier transform is a linear map between Hilbert spaces)
2. For $f \in C^\infty(\mathbb{T}^n)$ we have,

$$\begin{aligned} \widehat{\partial_{x_j} f}(k) &= \int_{\mathbb{T}^n} \partial_x f(x) e^{-i2\pi(k\cdot x)}\ dx \\ &= i2\pi k_j \int_{\mathbb{T}^n} f(x) e^{-i2\pi(k\cdot x)}\ dx \end{aligned}$$

   This implies that,

$$\widehat{\Delta f}(k) = -(2\pi)^2 |k|^2 \hat{f}(k)$$

   So essentially the Laplacian $\Delta \leq 0$

## 3.6 Some Notation

### 3.6.1 Sim

We write,

$$x \lesssim_a y$$

to mean,

$$x \leq C(a) y$$

where $C(a)$ is a constant depending on $a$.

We write,

$$x \sim_a y$$

When we have both $x \lesssim_a y$ and $y \lesssim_a x$

**Example.** If we have $a, b \geq 0$ then,

$$(a+b)^2 \sim a^2 + b^2$$

As we have $a^2 + b^2 \leq (a+b)^2 \leq 2(a^2+b^2)$ ◇

### 3.6.2 Japanese bracket

Let $x \in \mathbb{C}^n$ then, $\langle x \rangle = \sqrt{1 + \|x\|^2}$. So note that for $\|x\| > 1$ we have $\langle x \rangle \sim |x|$ as,

$$|x| \leq \sqrt{1 + |x|^2} \leq \sqrt{2}|x|$$

And for $|x| < 1$ we have $\langle x \rangle \sim 1$. Also we have $x \to \langle x \rangle$ is <u>smooth</u>.

## 3.7 Test Functions

$D(\mathbb{T}^n) = C^\infty(\mathbb{T}^n) = \bigcap_{k\geq 1} C^k(\mathbb{T}^n)$

Note that in this case $D(\mathbb{T}^n)$ is not a normed vector space, it's a Frechet space. But we say $v_n \to^{n\to\infty} v$ in $D(\mathbb{T}^n)$ when, $v_n \to v$ in $C^k(\mathbb{T}^n)\forall k$

A linear operator $\phi : D(\mathbb{T}^n) \to \mathbb{C}$ is <u>continuous</u> when,

$$\forall v_n \to^D v \Rightarrow \phi(v_n) \to^{\mathbb{C}} \phi(v)$$

$\phi$ is then called a <u>distribution</u> (a generalized function). We define $D'$ the space of distributions and hence $\phi \in D'$.

**Example.** Let $f \in C(\mathbb{T}^n)$ then define $\phi_f$ as,

$$\phi_f(v) = \int_{\mathbb{T}^n} fv \; \forall v \in D$$

here we can identify $f$ with $\phi_f$. In this case we have, $C(\mathbb{T}^n) \hookrightarrow D'$ (a continuous injection, this means for any $f \in C(\mathbb{T}^n)$ we can associate it unify with a $\phi_f \in D'$) ⋄

Essentially we got,

$$D = C^\infty \hookrightarrow C \hookrightarrow L^2 \hookrightarrow D'$$

Note take some $f, \phi_f$ such that $f$ is $C^1(\mathbb{T})$ then,

$$\int_{\mathbb{T}^n} f'v = -\int_{\mathbb{T}^n} fv' = \phi_f(-v')$$

using integration by parts. But note that LHS is undefined if $f \notin C^1$. But the RHS is still defined. Point being we can still identify $f'$ with some function $v \to \phi_f(-v')$.

So $\forall v \in D, \forall \phi \in D'$ :,

$$\phi'(v) := \phi(-v')$$

so we can <u>differentiate</u> any distribution.

The dirac delta is $\delta_0(v) = v(0)\forall v \in D$ so we have $\delta_0 \in D'$

## 3.8 Sobolev functions

A distribution $\phi \in D'$ is in $L^2$ when, $\forall v \in D$ :,

$$|\phi(v)| \lesssim \|v\|_{L^2}$$

so $\phi$ extends to a bounded linear functional on $L^2$. This gives the representation theorem i.e,

Any continuous linear functional on $L^2$ can be <u>represented</u> by an $L^2$ function.

**Sobolev function** Let $f \in D'(\mathbb{T}^n)$ we say that $f \in H^m(\mathbb{T}^n)$ when,

$$\partial^\alpha f \in L^2 \; \forall |\alpha| \leq m$$

so $f, f', \dots f^{(m)} \in L^2$

Note that $H^m$ is a normed vector space,